Review : Powers of a matrix

Given a square matrix A and a positive integer k, we define

$$A^k = \underbrace{AA\cdots A}_{k \text{ times}}$$

- Note that the multiplications AA, AAA, ... make sense.

Example.

Suppose $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Then $A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

$$A^3 = (A^2)A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

$$A^4 = (A^3)A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$$

Transpose of A

Suppose A is an $m \times n$ matrix.

The transpose of A, which is denoted by A^T, is an $n \times m$ matrix whose rows are columns of A.

Example. Suppose $A = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 3 & -\frac{1}{2} \end{bmatrix}$. Then $A^T = \begin{bmatrix} 2 & 0 \\ -1 & 3 \\ 1 & -\frac{1}{2} \end{bmatrix}$.

Properties of transpose

- $(A^T)^T = A$
- Suppose A and B are both $m \times n$ matrices and c is a scalar. Then $(A + B)^T = A^T + B^T$ and $(cA)^T = cA^T$.
- Suppose A is $m \times n$ and B is $n \times p$. Then $(AB)^T = B^T A^T$

Remark. Note that $(AB)^T = A^T B^T$ is generally not true!

For example, if A is 2×3 and B is 3×5, then $(AB)^T$ is defined and is a 5×2 matrix, whereas $A^T B^T$ is not even defined!

$$\underbrace{\begin{bmatrix} * & * & * \\ * & * & * \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix}}_{B} = \underbrace{\begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \end{bmatrix}}_{AB}$$

$$\underbrace{\begin{bmatrix} * & * \\ * & * \\ * & * \end{bmatrix}}_{A^T} \underbrace{\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}}_{B^T} = \text{☹}$$

Inverse of a matrix

Suppose A is an $n \times n$ matrix.

- We say that A is invertible if there exists an $n \times n$ matrix C such that

$$AC = CA = \mathrm{I}_n.$$

- The matrix C, if it exists, is called the inverse of A, and is usually denoted by A^{-1}.
- If A is invertible, then it has a *unique* inverse.

Example. Suppose $A = \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}$. Then if we take $C = \begin{bmatrix} -1 & 3 \\ 1 & -2 \end{bmatrix}$ we will have

$$AC = \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

and $CA = \begin{bmatrix} -1 & 3 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

This means that A is invertible and C is the inverse of A.

Question. When does an $n \times n$ matrix A have an inverse?

Theorem (2×2 matrices)

Suppose $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then :

- If $ad - bc = 0$ then A is **not** invertible.
- If $ad - bc \neq 0$ then A is invertible and we have

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

- The quantity $ad - bc$ is called the determinant of A.

Example. The matrix $A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$ is not invertible because $1 \times 6 - 3 \times 2 = 6 - 6 = 0$.

Example. The matrix $A = \begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix}$ is invertible because $1 \times 2 - 3 \times 2 = -4 \neq 0$.

Moreover,

$$A^{-1} = \frac{1}{-4} \begin{bmatrix} 2 & -3 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{3}{4} \\ \frac{1}{2} & -\frac{1}{4} \end{bmatrix}$$

Example

Is $A = \begin{bmatrix} 4 & 3 \\ -\frac{1}{3} & 1 \end{bmatrix}$ invertible? If so, then compute A^{-1}.

Solution. Determinant of A is $(4 \times 1) - (3 \times (-\frac{1}{3})) = 4 - (-1) = 5 \neq 0$. Therefore A is invertible and

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 1 & -3 \\ \frac{1}{3} & 4 \end{bmatrix} = \begin{bmatrix} \frac{1}{5} & -\frac{3}{5} \\ \frac{1}{15} & \frac{4}{5} \end{bmatrix}$$

Then,

$$AA^{-1} = \begin{bmatrix} 4 & 3 \\ -\frac{1}{3} & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{5} & -\frac{3}{5} \\ \frac{1}{15} & \frac{4}{5} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

and

$$A^{-1}A = \begin{bmatrix} \frac{1}{5} & -\frac{3}{5} \\ \frac{1}{15} & \frac{4}{5} \end{bmatrix} \begin{bmatrix} 4 & 3 \\ -\frac{1}{3} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Basic properties of inverse

Theorem

1. Suppose A is an $n \times n$ matrix. If A is invertible, then A^{-1} is invertible, and

$$(A^{-1})^{-1} = A$$

Reason : $A^{-1}A = AA^{-1} = \mathrm{I}_n$, i.e., A is the inverse of A^{-1}.

2. If A and B are $n \times n$ invertible matrices, then AB is also invertible and we have

$$(AB)^{-1} = B^{-1}A^{-1}$$

Reason : $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = A\mathrm{I}_nA^{-1} = AA^{-1} = \mathrm{I}_n$.

3. If A is an $n \times n$ invertible matrix, then so is A^T and the inverse of A^T is the transpose of A^{-1}. That is,

$$(A^T)^{-1} = (A^{-1})^T$$

Reason : $A^T(A^{-1})^T = (A^{-1}A)^T = \mathrm{I}_n^T = \mathrm{I}_n$.

Calculating A^{-1} in general

- Suppose you are given an $n \times n$ matrix A.

Question. Is A invertible? If it is, how can we find its inverse?

Notation

Recall that I_n is the $n \times n$ identity matrix, i.e.,

$$\mathrm{I}_n = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$
$$\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \cdots \quad \mathbf{e}_n$$

- We denote the i-th column of I_n by $\mathbf{e}_i$. Therefore

$$\mathrm{I}_n = [\, \mathbf{e}_1 \;\; \mathbf{e}_2 \;\; \cdots \;\; \mathbf{e}_n \,]$$

- The i-th entry of $\mathbf{e}_i$ is 1, and the rest of its entries are 0.

Suppose A is invertible, i.e., we can find an $n \times n$ matrix A^{-1} such that $AA^{-1} = \mathrm{I}_n$.

Write A^{-1} as

$$A^{-1} = \begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_n \end{bmatrix}$$

We have $AA^{-1} = \begin{bmatrix} A\mathbf{x}_1 & A\mathbf{x}_2 & \ldots & A\mathbf{x}_n \end{bmatrix}$.

But:

$$AA^{-1} = \mathrm{I}_n \quad \Rightarrow \quad \begin{bmatrix} A\mathbf{x}_1 & A\mathbf{x}_2 & \ldots & A\mathbf{x}_n \end{bmatrix} = \begin{bmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \cdots & \mathbf{e}_n \end{bmatrix}$$

It follows that

$$A\mathbf{x}_1 = \mathbf{e}_1 \quad , \quad A\mathbf{x}_2 = \mathbf{e}_2 \quad , \quad \cdots \quad , \quad A\mathbf{x}_n = \mathbf{e}_n$$

To determine if A is invertible, and to calculate A^{-1}, we should solve the following n matrix equations :

$$A\mathbf{x}_1 = \mathbf{e}_1 \quad , \quad A\mathbf{x}_2 = \mathbf{e}_2 \quad , \quad \cdots \quad , \quad A\mathbf{x}_n = \mathbf{e}_n$$

- To determine if A is invertible, and to calculate A^{-1}, we should solve n matrix equations :

$$A\mathbf{x}_1 = \mathbf{e}_1 \quad , \quad A\mathbf{x}_2 = \mathbf{e}_2 \quad , \quad \cdots \quad , \quad A\mathbf{x}_n = \mathbf{e}_n$$

- This means we have to solve the linear systems with the following augmented matrices :

$$\left[\, A \,\middle|\, \mathbf{e}_1 \,\right] \quad , \quad \left[\, A \,\middle|\, \mathbf{e}_2 \,\right] \quad , \quad \cdots \quad , \quad \left[\, A \,\middle|\, \mathbf{e}_n \,\right]$$

- we can solve these linear systems simultaneously by finding the R.E.F. of

$$\left[\, A \,\middle|\, \mathbf{e}_1 \;\; \mathbf{e}_2 \;\; \cdots \;\; \mathbf{e}_n \,\right] = \left[\, A \,\middle|\, \mathrm{I}_n \,\right]$$

Algorithm for finding A^{-1}

Suppose A is an $n \times n$ matrix.

Row reduce the super augmented matrix $\left[\begin{array}{c|c} A & \mathrm{I}_n \end{array}\right]$ to obtain its R.E.F.

- If after row reduction we obtain a matrix of the form

$$\left[\begin{array}{c|c} \mathrm{I}_n & B \end{array}\right]$$

  then A is invertible and $B = A^{-1}$.

- Otherwise, A is not invertible.

Example

Is $A = \begin{bmatrix} 1 & 0 & -2 \\ 1 & 0 & -3 \\ 0 & 1 & 2 \end{bmatrix}$ invertible? If so, then find its inverse.

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 0 & -3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \leftrightsquigarrow R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \\ 0 & 0 & -1 & -1 & 1 & 0 \end{array}\right] \xrightarrow{R_3 \to -R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 & 0 \end{array}\right]$$

$$\xrightarrow[R_2 \to R_2 - 2R_3]{R_1 \to R_1 + 2R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -2 & 0 \\ 0 & 1 & 0 & -2 & 2 & 1 \\ 0 & 0 & 1 & 1 & -1 & 0 \end{array}\right] \qquad \textit{R.E.F.}$$

Checking the answer: $AA^{-1} = \begin{bmatrix} 1 & 0 & -2 \\ 1 & 0 & -3 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & -2 & 0 \\ -2 & 2 & 1 \\ 1 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ ✓

Similarly, we can check that $A^{-1}A = \mathrm{I}_3$.

Example

Is $A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 0 & 1 \\ 3 & 1 & 0 \end{bmatrix}$ invertible? If it is, then find its inverse.

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 & 1 & 0 \\ 3 & 1 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & -2 & 3 & -2 & 1 & 0 \\ 0 & -2 & 3 & -3 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & -2 & 3 & -2 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 1 \end{array}\right] \xrightarrow{R_3 \to -R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & -2 & 3 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 \end{array}\right]$$

$$\xrightarrow[R_1 \to R_1 - R_3]{R_2 \to R_2 + 2R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & -1 & 1 \\ 0 & -2 & 3 & 0 & 3 & -2 \\ 0 & 0 & 0 & 1 & 1 & -1 \end{array}\right] \xrightarrow{R_2 \to -\frac{1}{2}R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & -1 & 1 \\ 0 & 1 & -\frac{3}{2} & 0 & -\frac{3}{2} & 1 \\ 0 & 0 & 0 & 1 & 1 & -1 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 1 & -\frac{3}{2} & 0 & -\frac{3}{2} & 1 \\ 0 & 0 & 0 & 1 & 1 & -1 \end{array}\right] \quad \textit{R.E.F.}$$

It follows that A is <u>not</u> invertible.

Example

Is $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}$ invertible? If it is, then find its inverse.

Solution.

$$\left[\begin{array}{ccc|ccc} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \leftrightsquigarrow R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - 4R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -3 & -4 & 0 & -4 & 1 \end{array}\right] \xrightarrow{R_3 \to R_3 + 3R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 3 & -4 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to \frac{1}{2} R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \end{array}\right] \xrightarrow[R_1 \to R_1 - 3R_3]{R_2 \to R_2 - 2R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{9}{2} & 7 & -\frac{3}{2} \\ 0 & 1 & 0 & -2 & 4 & -1 \\ 0 & 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \end{array}\right]$$

It follows that A is invertible and $A^{-1} = \begin{bmatrix} -\frac{9}{2} & 7 & -\frac{3}{2} \\ -2 & 4 & -1 \\ \frac{3}{2} & -2 & \frac{1}{2} \end{bmatrix}$.

Criteria for invertibility

Theorem

Let A be an $n \times n$ matrix. Then the following statements are equivalent (i.e., they are either all true or all false).

1. A is an invertible matrix.
2. The R.E.F. of A is I_n.
3. A has n pivot positions.
4. The equation $A\mathbf{x} = \mathbf{0}$ only has the trivial solution $\mathbf{x} = \mathbf{0}$.
5. The columns of A are linearly independent.
6. For every vector $\mathbf{b}$ in $\mathbb{R}^n$, the matrix equation $A\mathbf{x} = \mathbf{b}$ has a unique solution.
7. The columns of A span $\mathbb{R}^n$.
8. A^T is invertible.

Example

Suppose that A is an $n \times n$ matrix which satisfies

$$A^5 - 2A^3 + A - 2\mathrm{I} = 0.$$

Is A invertible?

Solution. One can write

$$A\left(A^4 - 2A^2 + \mathrm{I}\right) = 2\mathrm{I}$$

or

$$A\left(\frac{1}{2}A^4 - A^2 + \frac{1}{2}\mathrm{I}\right) = \left(\frac{1}{2}A^4 - A^2 + \frac{1}{2}\mathrm{I}\right)A = \mathrm{I}$$

It follows that A is invertible.
In fact the above relation shows that :

$$A^{-1} = \frac{1}{2}A^4 - A^2 + \frac{1}{2}\mathrm{I}$$

Matrix Equations

Example

Solve the equation $AXB = C$ for X where

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 1 & 0 & -3 \\ 0 & 1 & 2 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & -2 \\ 0 & -1 \end{bmatrix} \qquad \text{and } C = \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ -3 & 3 \end{bmatrix}$$

Solution. First of all, note that since A is 3×3 and B is 2×2, X should be 3×2.
If A and B were invertible then we could write :

$$A^{-1}AXBB^{-1} = A^{-1}CB^{-1} \quad \Longrightarrow \quad \mathrm{I}_3 X \mathrm{I}_2 = A^{-1}CB^{-1} \quad \Longrightarrow X = A^{-1}CB^{-1}$$

We now check to see if A and B indeed invertible. For A, we have

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 0 & -3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -2 & 0 \\ 0 & 1 & 0 & -2 & 2 & 1 \\ 0 & 0 & 1 & 1 & -1 & 0 \end{array}\right]$$

Which means that A is invertible and $A^{-1} = \begin{bmatrix} 3 & -2 & 0 \\ -2 & 2 & 1 \\ 1 & -1 & 0 \end{bmatrix}$.

Example

Solve the equation $AXB = C$ for X where

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 1 & 0 & -3 \\ 0 & 1 & 2 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & -2 \\ 0 & -1 \end{bmatrix} \quad \text{and } C = \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ -3 & 3 \end{bmatrix}$$

For $B = \begin{bmatrix} 1 & -2 \\ 0 & -1 \end{bmatrix}$, we can use the method of this lecture or the method of the last lecture :

determinant of B is $-1 \neq 0$, hence $B^{-1} = \frac{1}{-1}\begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 0 & -1 \end{bmatrix}$

Finally, we have

$$X = A^{-1}CB^{-1} = \begin{bmatrix} 3 & -2 & 0 \\ -2 & 2 & 1 \\ 1 & -1 & 0 \end{bmatrix}\begin{bmatrix} 1 & -2 \\ 0 & 1 \\ -3 & 3 \end{bmatrix}\begin{bmatrix} 1 & -2 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & -8 \\ -5 & 9 \\ 1 & -3 \end{bmatrix}\begin{bmatrix} 1 & -2 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ -5 & 1 \\ 1 & 1 \end{bmatrix}$$

Example

Let $A = \begin{bmatrix} 1 & 3 \\ -4 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 100 \\ 1 & -1 \end{bmatrix}$. Solve the equation

$$B(X^T + A)B^{-1} = 2\mathrm{I} \qquad \mathrm{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

for X.

Solution. Since A and B are 2×2, the matrix X is also 2×2. (We also have $\mathrm{I} = \mathrm{I}_2$.)

$$B(X^T + A)B^{-1} = 2\mathrm{I} \;\Rightarrow\; B^{-1}B(X^T + A)B^{-1} = 2B^{-1}\mathrm{I} = 2B^{-1}$$

$$\Rightarrow \quad (X^T + A)B^{-1} = 2B^{-1}$$

$$\Rightarrow \quad (X^T + A)B^{-1}B = 2B^{-1}B = 2\mathrm{I} \;\Rightarrow\; X^T + A = 2\mathrm{I}$$

$$\Rightarrow X^T = 2\mathrm{I} - A$$

$$\Rightarrow \quad X = (2\mathrm{I} - A)^T = 2\mathrm{I} - A^T = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 1 & -4 \\ 3 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ -3 & -3 \end{bmatrix}.$$

Example

Simplify $A^{-1}(BC)^T(B^{-1})^T(C^{-1})^T A^2$

Solution. We have

$$\begin{aligned}
A^{-1}(BC)^T(B^{-1})^T(C^{-1})^T A^2 &= A^{-1}C^T B^T(B^{-1})^T(C^{-1})^T A^2 \\
&= A^{-1}C^T\left(B^T(B^{-1})^T\right)(C^{-1})^T A^2 \\
&= A^{-1}C^T\left(B^T(B^T)^{-1}\right)(C^{-1})^T A^2 \\
&= A^{-1}C^T\,\mathrm{I}\,(C^{-1})^T A^2 \\
&= A^{-1}C^T(C^{-1})^T A^2 \\
&= A^{-1}C^T(C^T)^{-1}A^2 \\
&= A^{-1}\,\mathrm{I}\,A^2 \\
&= A^{-1}A^2 \\
&= A
\end{aligned}$$

Example

Suppose $A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$ and B is a 2×3 matrix such that $AB = \begin{bmatrix} 1 & -1 & 2 \\ -3 & 2 & 1 \end{bmatrix}$
What is the second column of B?

Solution. If we write B as $B = [\ \mathbf{b}_1\ \ \mathbf{b}_2\ \ \mathbf{b}_3\ ]$ then we have

$$AB = [\ A\mathbf{b}_1\ \ A\mathbf{b}_2\ \ A\mathbf{b}_3\ ]$$

And therefore

$$A\mathbf{b}_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

But determinant of A is $2 \neq 0$, hence A is invertible and by the formula from the last lecture we have

$$A^{-1} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \qquad \Longrightarrow \mathbf{b}_2 = A^{-1}\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -\frac{3}{2} \\ \frac{1}{2} \end{bmatrix}$$

Today's key point : Algorithm for finding A^{-1}

Suppose A is an $n \times n$ matrix.

Row reduce the super augmented matrix $\left[\begin{array}{c|c} A & \mathrm{I}_n \end{array}\right]$ to obtain its R.E.F.

- If after row reduction we obtain a matrix of the form

$$\left[\begin{array}{c|c} \mathrm{I}_n & B \end{array}\right]$$

  then A is invertible and $B = A^{-1}$.

- Otherwise, A is not invertible.